National Tsing Hua University
Department of Electrical Engineering
EE6620 Computational Photography, Spring 2026

Homework #1
High Dynamic Range Imaging

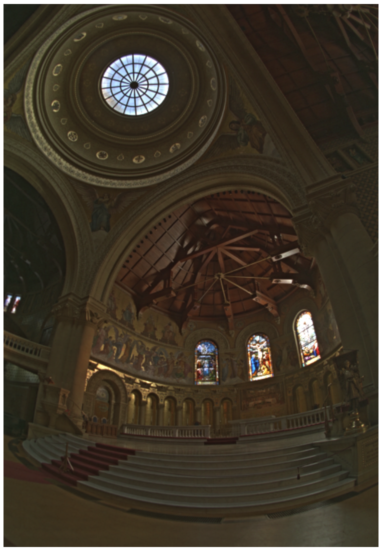
Assigned on March 11, 2026

Due by April 1, 2026

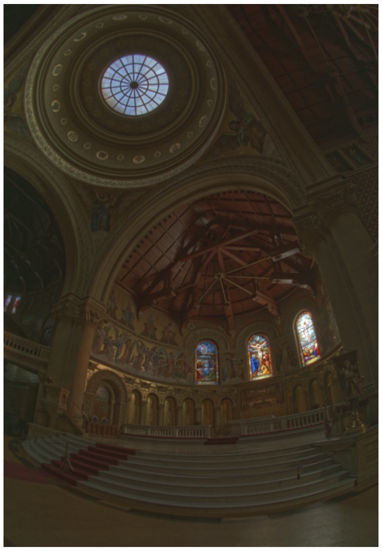
Homework Description

Modern cameras are unable to capture the full dynamic range of commonly encountered natural scenes. High-dynamic-range (HDR) photographs are generally achieved by capturing multiple standard-exposure images, often using exposure bracketing, and then merging them into a single HDR image. Also, to view the HDR image on an ordinary low-dynamic-range (LDR) display, tone mapping operation from HDR to LDR on images is required. In this assignment, you will implement the whole HDR photography flow, including **image bracketing**, **camera response calibration**, **white balance**, and finally, **tone mapping**, to visualize your results.

GlobalTM



LocalTM (Gaussian)



LocalTM (Bilateral)

Table 1: Scores

Items		Score
Implementation (30%)	Camera Response Calibration	10%
	White Balance	5%
	Global Tone Mapping	5%
	Local Tone Mapping (Gaussian)	5%
	Local Tone Mapping (Bilateral)	5%
Report (70%)	Exploration	25%
	Research Study	45%

For coding environment, we are using Google Colab. More information can be seen in Python_Tutorial.pdf. You must submit through sharing the entire folder with `minaminamay30@gmail.com` as editor. Do not download and re-upload since coding histories would be checked. **Any work lacking editing history will be flagged for plagiarism or AI generation, leading to a 25% grade penalty.**

Note that in implementation part, you can only use the basic functions for array operations in numpy, e.g., np.mean(), np.sum(), np.exp(), np.pad(), etc. **You cannot use functions in packages for image processing or code acceleration, e.g., cv2.GaussianBlur(), cv2.bilateralFilter(), cffi.FFI(), etc.**

1 Implementation

In this section, you will implement the algorithms for the whole HDR imaging flow. For each function you need to implement in HDR_functions.ipynb, the corresponding test function is provided in test_HDR_functions.ipynb to help you check the correctness. You can develop your functions step by step with these unit tests to complete the whole HDR flow. In test_HDR_functions.ipynb, argument TEST_PAT_SIZE can be assigned to either `small` or `large`. `small` stands for small test pattern size and `large` stands for large test pattern size. It is recommended to set TEST_PAT_SIZE `small` for quick debugging during implementation. However, you have to **pass the unit test with TEST_PAT_SIZE** `large` **to get the full score in each part. And please do not modify original function IO. Otherwise, your implementation will be considered as incorrect.** Note that for large pattern size, the bilateral filtering process may take longer time to complete. **If the hold execution time (8 validation functions in test_HDR_functions.ipynb) of your code exceeds 2 minutes, please optimize your functions; otherwise, you will get 5% punishment.** One test image, memorial, is provided in folder /TestImage. After passing all unit tests, you can execute HDR_flow.ipynb to run the whole HDR imaging flow.

Figure 1: HDR imaging flow

1.1 Camera Response Calibration (10%)

Camera response calibration estimates the radiometric response of a camera. In this assignment, you will implement the debevec method [1] to recover a high dynamic range radiance map from a set of bracketing images.

The debevec method consists of 3 steps:

a. Describe the objective function in matrix form.

$$O = \sum_{i}\sum_{j}[w(Z_{ij})(g(Z_{ij}) - \ln E_i - \ln \Delta t_j)]^2 + \lambda \sum_{z=Z_{min}+1}^{Z_{max}-1} [w(z)g''(z)]^2 \tag{1}$$

In the objective function, Z_{ij} is the intensity measurement for pixel i on image j. Δt_j is the exposure time of the image j, and E_i is our target radiance of location i. $g(z)$ is the camera response which maps the measured intensity Z_{ij} to exposure $E_i \Delta t_j$. The objective function connects data term and smoothness term with a Lagrange multiplier λ. A curve shape prior $w(z)$ is applied in debevec method to fit the curve assumption:

$$w(z) = \begin{cases} z - Z_{min} & z \leq \frac{1}{2}(Z_{min} + Z_{max}) \\ Z_{max} - z & z > \frac{1}{2}(Z_{min} + Z_{max}) \end{cases} \tag{2}$$

Besides, we should also apply an unit exposure assumption by letting $g(127) = 0$. We will describe the equation and prior in the matrix form $Ax = b$. The relation between the objective function and the matrix is shown in Figure 2.

For data term, we would like to fill the matrix with

$$[w(Z_{ij}),\ -w(Z_{ij})] \cdot [g(Z_{ij}),\ \ln E_i] = w(Z_{ij}) \ln \Delta t_j \tag{3}$$

for measurement $(Z_{ij}, \Delta t_j)$.

And for smoothness term with Lagrange multiplier, we would like to fill the matrix with

$$\lambda[w(z),\ -2w(z),\ w(z)] \cdot [g(z-1),\ g(z),\ g(z+1)] = 0 \tag{4}$$

for intensity z.

b. Solve the least square problem with function `numpy.linalg.lstsq()`.

c. Estimate the radiance map from response curve. The final radiance for each pixel i can be estimated by averaging measurements from all images j:

$$\ln E_i = \frac{\sum_j w(Z_{ij})(g(Z_{ij}) - \ln \Delta t_j)}{\sum_j w(Z_{ij})} \tag{5}$$

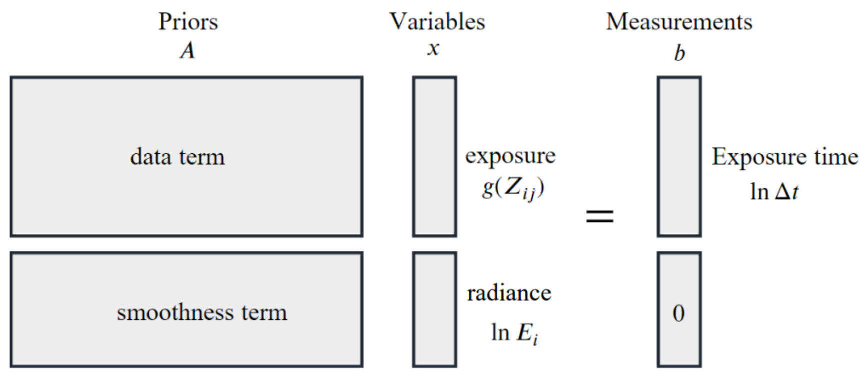


Figure 2: Matrix Form

Complete functions `prior()` (Eq. (5)), `Response_Estimation()` (steps a, b) and `Radiance_Construction()` (step c) in HDR_functions.ipynb.

1.2 White balance (5%)

White balance is the process of removing unrealistic color casts, so that the objects which appear white in person are rendered white in your photo/display. In this part, the color ratios are scaled such that the sum of intensities across the three color channels are equal within "known to be white" region. To achieve this, color channels R and G are scaled with B_{sum}/R_{sum} and B_{sum}/G_{sum} respectively based on the sum value measured in the "known to be white" region:

$$X_{sum} = \text{sum}_{(i,j)\in\Omega_{ktbw}} X \tag{6}$$

$$X' = X * (B_{sum}/X_{sum}),\ for\ X \in \{R, G\} \tag{7}$$

R, G, and R', G' here stand for the radiance before and after the white balance.
Complete function `White_Balance()` in HDR_functions.ipynb.

1.3 Global tone mapping (5%)

Global tone mapping compresses the contrast of intensity by scaling in log domain as shown in Eq. (8)

$$\log_2 \hat{X} = s(\log_2 X - \log_2 X_{max}) + \log_2 X_{max}, \tag{8}$$

$$X_{max} = \max_{i,j}(X(i,j))$$

where X is the radiance, and $\hat{X}$ is the compressed result. Note that this operation should be performed on each of the color channels. Besides the tone mapping operation, we should also perform gamma correction

$$X' = \hat{X}^{1/\gamma} \tag{9}$$

to convert linear RGB to nonlinear color space and transform range to 0-255 before saving.
Steps to transform to 0-255 domain: clip to range[0,1] $\rightarrow$ multiply by 255 $\rightarrow$ round to integers.
("clip to range[0,1]" means if the value$<$0, assign it to 0; if the value$>$1, assign it to 1.)

In tone mapping, the contrast is reduced in log domain and s here is a parameter to adjust the contrast. On the other hand, γ in nonlinear correction is fixed to commonly used 2.2.
Complete function `Global_Tone_Mapping()` in HDR_functions.ipynb.

1.4 Local tone mapping with Gaussian filter (5%)

Local tone mapping performs contrast compression only on the base layer to preserve details while balancing the luminance of different regions. A reference flow to achieve this goal is listed as below. Note that you should use **symmetric padding** in the filtering.

Local Tone Mapping Flow:

a. Separate Intensity map(I) and Color Ratio(C_r, C_g, C_b) for radiance (R, G, B).

$$I = \text{avg}(R, G, B) \tag{10}$$

$$C_x = X/I,\ for\ X \in \{R, G, B\} \tag{11}$$

b. Take log of intensity

$$L = \log_2 I \tag{12}$$

c. Separate the detail layer (L_D) and base layer (L_B) of L with Gaussian filter,

$$L_B(i,j) = \frac{\sum_{k,l} L(i+k, j+l)w(k,l)}{\sum_{k,l} w(k,l)}, \tag{13}$$

$$w_{Gaussian}(k,l) = \exp\left(-\frac{||(0,0)-(k,l)||^2}{2\sigma_s^2}\right),\ k,l \in [-N/2, N/2] \tag{14}$$

$$L_D = L - L_B \tag{15}$$

Eq. (13) express the convolution between an image L and symmetric filter w. Indices i, j indexes the pixel of an image on location (i, j). Indices k, l span a filter with $N \times N$ window size. σ_s^2 in Eq. (14) is the variance of the Gaussian filter.

d. Compress the contrast

$$L'_B = (L_B - L_{max}) * \frac{scale}{L_{max} - L_{min}}, \tag{16}$$

$$L_{min} = \min_{i,j} L_B(i,j),$$

$$L_{max} = \max_{i,j} L_B(i,j)$$

where $scale$ is a parameter for contrast adjustment, different scenes may have different settings. The value around 2~15 would look good for most of scenes.

e. Reconstruct intensity map with adjusted base layer and detail layer,

$$I' = 2^{L'_B + L_D} \tag{17}$$

f. Reconstruct color map with adjusted intensity and color ratio,

$$C = C_x * I',\ for\ X \in \{R, G, B\} \tag{18}$$

Besides the tone mapping operation, we should also perform gamma correction Eq. (9) to convert linear RGB to nonlinear color space and transform range to 0-255 before saving.

Complete functions `Local_Tone_Mapping()` and `Gaussian_Filter()` in HDR_functions.ipynb.

1.5 Local tone mapping with Bilateral filter (5%)

Replace Gaussian filter in local tone mapping (sec 1.4 step c) with bilateral filter.

Bilateral filter on image L can be expressed as

$$w_{bilateral}(L, i, j, k, l) = \exp\left(-\frac{||(0,0) - (k,l)||^2}{2\sigma_s^2} - \frac{||L(i,j) - L(i+k, j+l)||^2}{2\sigma_r^2}\right) \tag{19}$$

where σ_s^2 and σ_r^2 represent the spatial and range variances of the bilateral filter. Again, indices k, l are bounded in $[-N/2, N/2]$ window.

Complete function `Bilateral_Filter()` in HDR_functions.ipynb.

2 Report (70%)

In this section, we will go into detail over the HDR imaging flow. There are two parts of problems: Exploration and Research Study. Answer the problems clearly and integrally in a report. **Note that you must type in Google Docs and install "Process Feedback" as Chrome Extensions.** More information can be seen in Python_Tutorial.pdf.

2.1 Exploration (25%)

a. A conceptual figure Figure 3 shows how bracketing images can recover the radiance of a scene. What is the meaning of this figure? Try to plot the same figure for memorial image set and describe how you plot the figure. **(5%)**

b. At the step of local tone mapping, why bilateral filter can eliminate halo effect, but gaussian filter cannot? **(5%)**

c. Create your own HDR image. Photograph a bracketing set of exposures for a designed scene and perform the HDR imaging flow. Generate at least three HDR images using different tone mapping methodology in the flow. i.e. Run the flow with global tone mapping, local tone mapping with gaussian filter, and local tone mapping with bilateral filter. Provide at least the following information in your report. **(15%)**

   i. Your bracketing images and result HDR images
   ii. How do you choose the scene? Note that not every scene needs to perform HDR imaging flow.
   iii. What are your settings of camera when photographing?
   e.g. exposure time, shutter speed, f-stop, . . .
   iv. How do you decide the parameters in the HDR imaging flow?

If you apply any additional steps, also give some description in the report. Additionally, if your initial attempt at creating an image was unsuccessful, include the failed versions in your report and explain why they did not work, as well as how a better result could be achieved.

Note that using pre-exist datasets from the internet is prohibited. Instead, capture your own bracketing images for the HDR imaging process. You can utilize items such as a flashlight or desk lamp to create a suitable HDR scene. Your student ID card must be presented in the images or no points would be given.

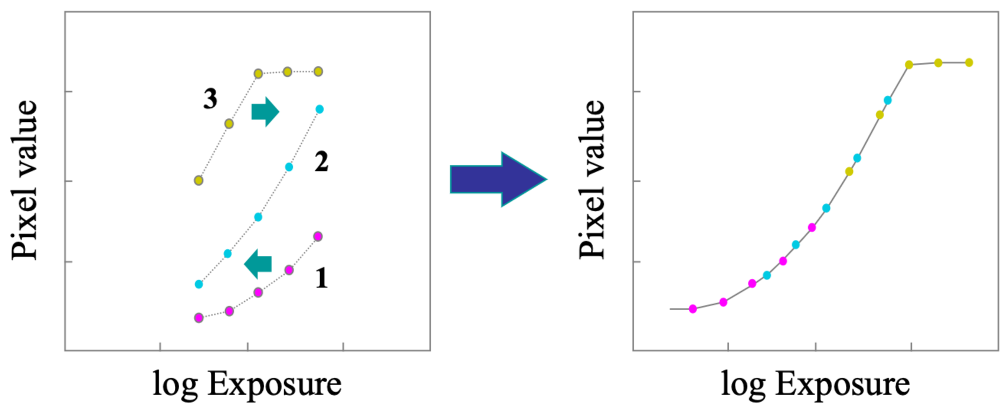


Figure 3: Radiometric Calibration

2.2 Research Study (45%)

In this part, you should answer the problems by the two-step research flow: **Assumption** and **Justification**. Based on your observation, propose an assumption to the question, and justify your ideas by experiments. The research process may be a loop:

Observation → Propose Assumption → Justification → Modify Assumption → Justification → ...

After you reach a satisfying result, clearly describe your assumption and how you justify it by experiments in the report. Additionally, you must record the entire research flow (including the failed methods and their experiments). For assumption part, you have to provide reasons explaining why you think so. For justification part, clearly describe how you design the experiment, whether the experiment results match your idea or not, and why the experiment results can justify your ideas. Grading criteria for this part are based on clarity of the report, rigorousness of the justification and the overall research flow.

For the first problem in this part, we will provide some guidance to help you understand this problem format. If you have no idea how to start, you can refer to the guidance. It may give you some ideas about the problem. Note that it is not mandatory to follow the guidance. You can do the research and describe it in your way.

a. In the objective function Eq. (1), the author apply a response curve assumption: the **smoothness term** which constrains the curve shape. Propose your idea to explain why this assumption reasonable and what will happen if we ignore the assumption. Then justify your ideas by experiments. **(15%)**

   Guidance:

   Assumption

   During camera response calibration, we will try to minimize the objective function. And in the objective function, the smoothness term contains $g''(z)$. By the definition or the meaning of the second order differential, ... Considering the relation between exposure and pixel value, ...

   Justification

   Plot and compare the conceptual figure with and without the smoothness term. See if the difference match the assumption.

b. In the tone mapping parts, following the application of gamma correction to convert linear RGB to a nonlinear color space, the data is further transformed into the 0-255 domain through a series of steps: clipping to the range [0,1] $\rightarrow$ multiplication by 255 $\rightarrow$ rounding to integers. Note that during the "clip to [0,1]" step, numerous pixels may become saturated to 1, potentially resulting in bad visual outcomes.

   Are there alternative methods for transforming the data into the 0-255 domain that might yield better results? Suggest your own algorithm. (Hint: Consider adjusting values based on the data's distribution rather than directly clipping to [0,1].) You have to do your experiments on both the provided dataset (memorial) and your own HDR images, which may require different methods. **(15%)**

c. When creating your HDR image from multiple exposures in **section 2.1.c**, objects that move between captures or camera shake caused by hand movement often result in "ghost artifacts". Please create at least one scenario that causes ghost artifacts (for example, by intentionally introducing object motion or hand shake). **Your student ID card must be presented in the images or no points would be given.**

Then, propose an assumption regarding how to eliminate ghost artifacts during the HDR imaging flow, and design experiments to justify or refute that assumption. The four required pieces of information in **section 2.1.c** should also be provided. (Note that no matter what method you use, please provide a detailed explanation of your motivation, idea, assumption, and justification.) **(15%)**

3 Deliverable

1. Submit only your report with filename {studentID}_report.pdf to eeclass.
2. Modify your folder name to HW1_{studentID}, and share with `minaminamay30@gmail.com` as editor. (Folder structure shall be identical to the assigned homework.)
   (a) Your HDR_functions.ipynb and any .ipynb written for the report in folder /code.
   (b) Your result HDR images of problem 2.1.c and 2.2.c in folder /MyHDR_result
   (c) Your bracketing images in folder /TestImage
   (d) Your report in Google Doc

Note that editing histories will be reviewed. As mentioned earlier, if no history is found, the work may be considered generated by AI or plagiarized, resulting in a 25% penalty. Additionally, wrong file delivery or arrangement will get 5% punishment.

References

[1] Debevec, Paul E., and Jitendra Malik. "Recovering high dynamic range radiance maps from photographs." *ACM SIGGRAPH 2008 classes* (2008): 1-10.